

General Purpose Transistor

Medium Power, PNP

80 V, 1 A

BCP53M

The BCP53MTW is designed for general purpose amplifier applications. It is housed in DFN2020–3 offering superior thermal performance. The transistor is ideal for medium–power surface mount applications where board space and reliability are at a premium.

Specification Features

- Wettable Flank Package for Optimal Automated Optical Inspection (AOI)
- NSV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC–Q101 Qualified and PPAP Capable
- These Devices are Pb–Free, Halogen Free/BFR Free and are RoHS Compliant

MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$)

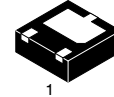
Rating	Symbol	Max	Unit
Collector–Emitter Voltage	V_{CEO}	–80	Vdc
Collector–Base Voltage	V_{CBO}	–100	Vdc
Emitter–Base Voltage	V_{EBO}	–6.0	Vdc
Collector Current – Continuous (Note 1)	I_C	1.0	A
Collector Current – Peak (Note 1)	I_{CM}	2.0	A

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

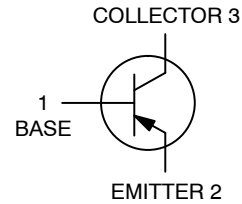
THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Total Power Dissipation (Note 2) @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5	W
Thermal Resistance, Junction–to–Ambient (Note 2)	$R_{\theta JA}$	78	$^\circ\text{C/W}$
Total Power Dissipation (Note 3) @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	875	mW
Thermal Resistance, Junction–to–Ambient (Note 3)	$R_{\theta JA}$	138	$^\circ\text{C/W}$
Junction and Storage Temperature Range	T_J, T_{stg}	–65 to +150	$^\circ\text{C}$

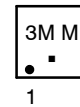
1. Reference SOA Curve
2. Surface–mounted on FR4 board using a 600 mm² pad area and 2 oz. Cu
3. Surface–mounted on FR4 board using a 100 mm² pad area and 2 oz. Cu



WDFNW3
CASE 515AA



MARKING DIAGRAM



3M = Specific Device Code
M = Date Code

ORDERING INFORMATION

See detailed ordering, marking and shipping information in the package dimensions section on page 5 of this data sheet.

BCP53M

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristics	Symbol	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

Collector-Emitter Breakdown Voltage ($I_C = -1\text{ mA}$, $I_B = 0\text{ A}$)	$V_{(BR)CEO}$	-80	-	-	V
Collector-Base Breakdown Voltage ($I_C = -100\text{ }\mu\text{A}$, $I_E = 0\text{ A}$)	$V_{(BR)CBO}$	-100	-	-	V
Emitter-Base Breakdown Voltage ($I_E = -10\text{ }\mu\text{A}$, $I_C = 0$)	$V_{(BR)EBO}$	-5	-	-	V
Collector-Base Cutoff Current ($V_{CB} = -30\text{ V}$, $I_E = 0$)	I_{CBO}	-	-	-100	nA
Emitter-Base Cutoff Current ($V_{EB} = -5\text{ V}$, $I_C = 0$)	I_{EBO}	-	-	-100	nA

ON CHARACTERISTICS (Note 4)

DC Current Gain ($I_C = -5\text{ mA}$, $V_{CE} = -2.0\text{ V}$) ($I_C = -150\text{ mA}$, $V_{CE} = -2.0\text{ V}$) ($I_C = -500\text{ mA}$, $V_{CE} = -2.0\text{ V}$)	All Part Types BCP53M BCP5310M BCP5316M All Part Types	h_{FE}	63 63 63 100 40	- - - - -	- 250 160 250 -	
Collector-Emitter Saturation Voltage ($I_C = -500\text{ mA}$, $I_B = -50\text{ mA}$)	$V_{CE(sat)}$	-	-	-	-0.50	V
Base-Emitter Saturation Voltage ($I_C = -500\text{ mA}$, $I_B = -50\text{ mA}$)	$V_{BE(sat)}$	-	-	-	-2.0	V
Base-Emitter Turn-on Voltage ($I_C = -500\text{ mA}$, $V_{CE} = -2.0\text{ V}$)	$V_{BE(on)}$	-	-	-	-1.0	V

SMALL SIGNAL CHARACTERISTICS

Transition Frequency ($I_C = -50\text{ mA}$, $V_{CE} = -5.0\text{ V}$, $f = 100\text{ MHz}$)	f_T	-	130	-	-	MHz
Output Capacitance ($V_{CB} = 10\text{ V}$, $f = 1.0\text{ MHz}$)	C_{obo}	-	12	-	-	pF
Input Capacitance ($V_{EB} = -0.5\text{ Vdc}$, $I_C = 0$, $f = 1.0\text{ MHz}$)	C_{ibo}	-	110	-	-	pF
Input Impedance ($I_C = -1.0\text{ mAdc}$, $V_{CE} = -10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{ie}	-	5	-	-	k
Voltage Feedback Ratio ($I_C = -1.0\text{ mAdc}$, $V_{CE} = -10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{re}	-	1	-	-	$\times 10^{-4}$
Small-Signal Current Gain ($I_C = -1.0\text{ mAdc}$, $V_{CE} = -10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{fe}	-	200	-	-	-
Output Admittance ($I_C = -1.0\text{ mAdc}$, $V_{CE} = -10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	H_{oe}	-	10	-	-	μmhos
Noise Figure ($I_C = 0.2\text{ mA}$, $V_{CE} = 5.0\text{ Vdc}$, $R_S = 2.0\text{ k}\Omega$, $f = 1.0\text{ kHz}$, $BW = 200\text{ Hz}$)	NF	-	0.8	-	-	dB

SWITCHING CHARACTERISTICS

Delay Time ($V_{CC} = 30\text{ Vdc}$, $I_C = 150\text{ mA}$, $I_{B1} = 15\text{ mA}$)	t_d	-	18	-	-	ns
Rise Time ($V_{CC} = 30\text{ Vdc}$, $I_C = 150\text{ mA}$, $I_{B1} = 15\text{ mA}$)	t_r	-	32	-	-	ns
Storage Time ($V_{CC} = 30\text{ Vdc}$, $I_C = 150\text{ mA}$, $I_{B1} = 15\text{ mA}$, $I_{B2} = 15\text{ mA}$)	t_s	-	660	-	-	ns
Fall Time ($V_{CC} = 30\text{ Vdc}$, $I_C = 150\text{ mA}$, $I_{B1} = 15\text{ mA}$, $I_{B2} = 15\text{ mA}$)	t_f	-	50	-	-	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. Pulse Condition: Pulse Width = 300 μs , Duty Cycle $\leq 2\%$.



TYPICAL CHARACTERISTICS

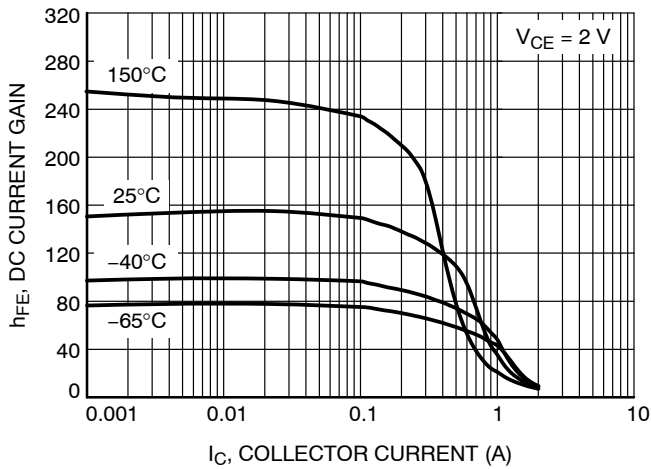


Figure 1. DC Current Gain

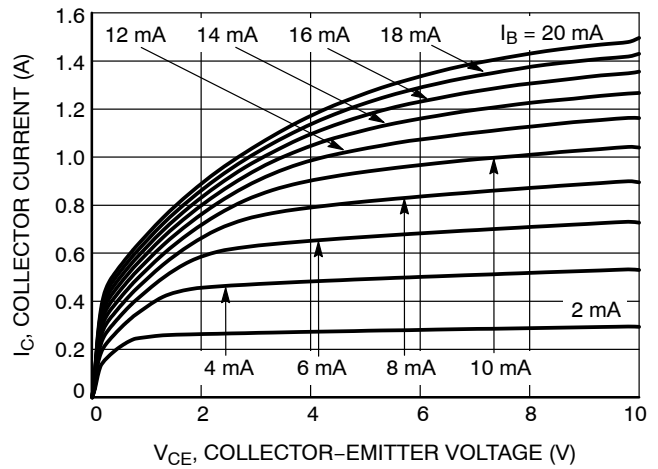


Figure 2. Collector Current vs. Collector Emitter Voltage

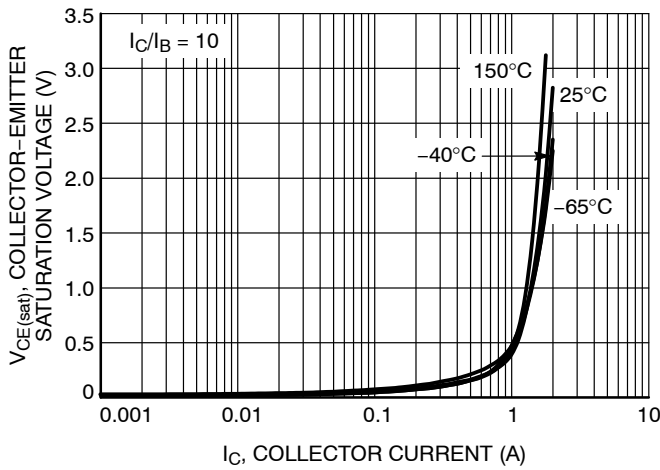


Figure 3. Collector Emitter Saturation Voltage vs. Collector Current

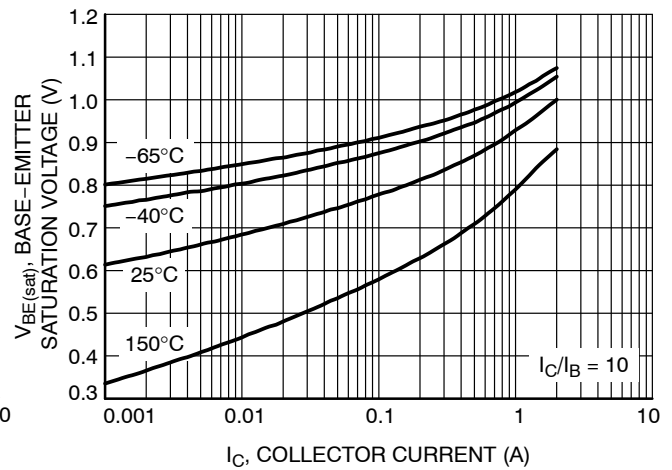


Figure 4. Base Emitter Saturation Voltage vs. Collector Current

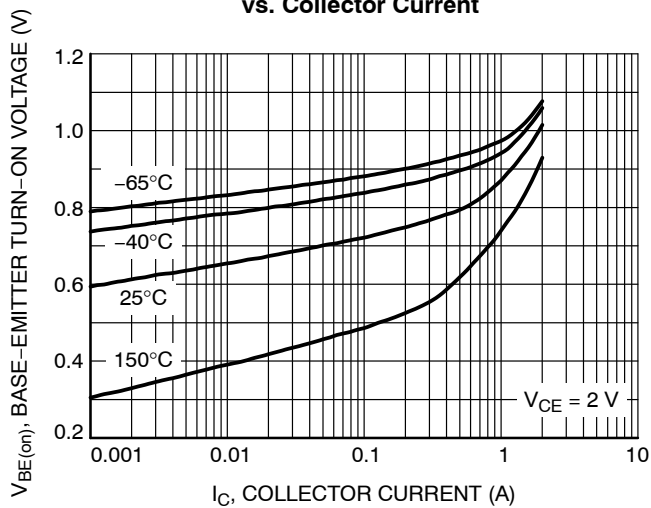
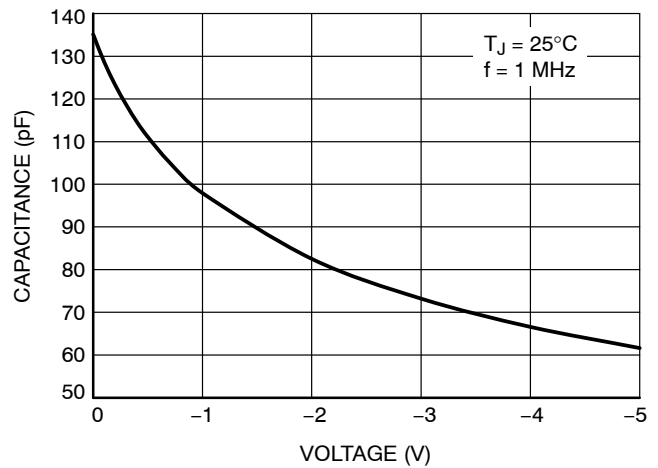
Figure 5. BCP53M, Base Emitter Turn-On Voltage vs. Collector Current $V_{BE(on)}$ 

Figure 6. Input Capacitance

TYPICAL CHARACTERISTICS

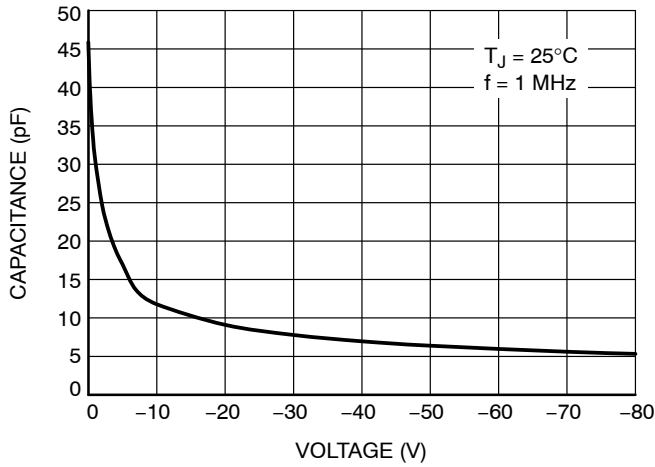


Figure 7. Output Capacitance

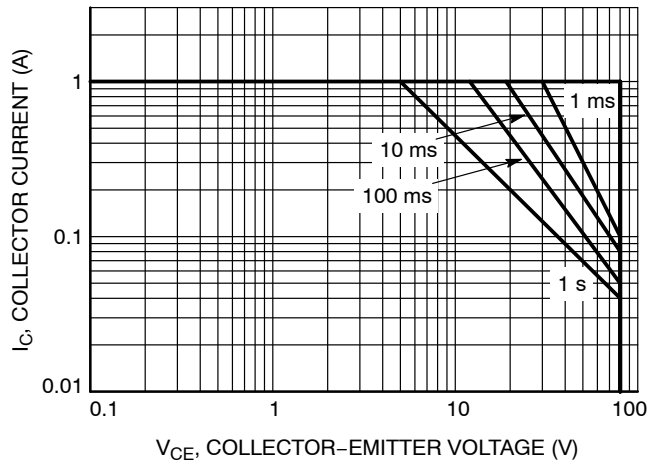


Figure 8. Safe Operating Area

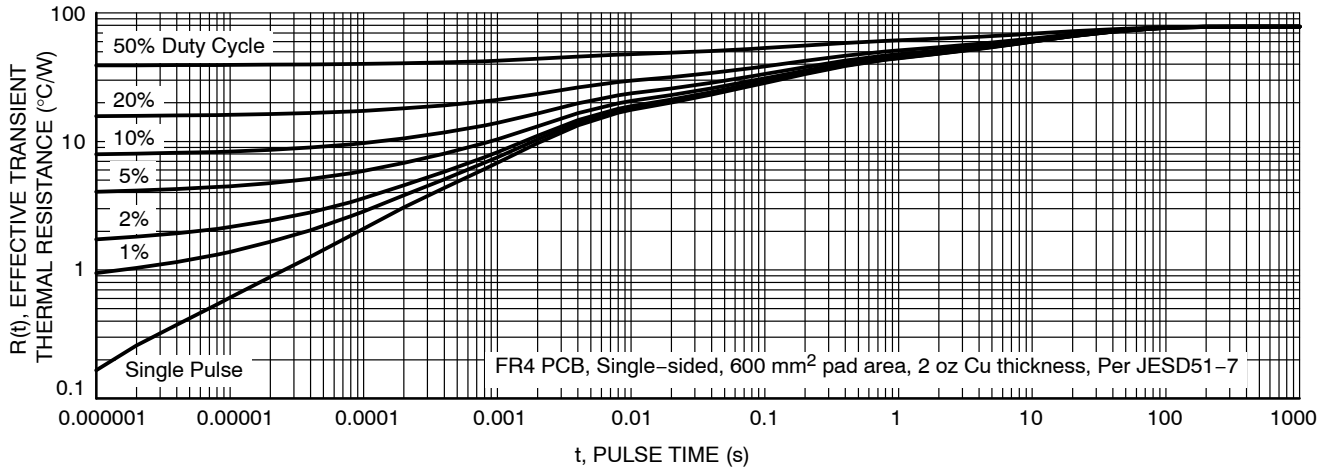


Figure 9. Transient Thermal Impedance from Junction-to-Ambient as a Function of Pulse Duration

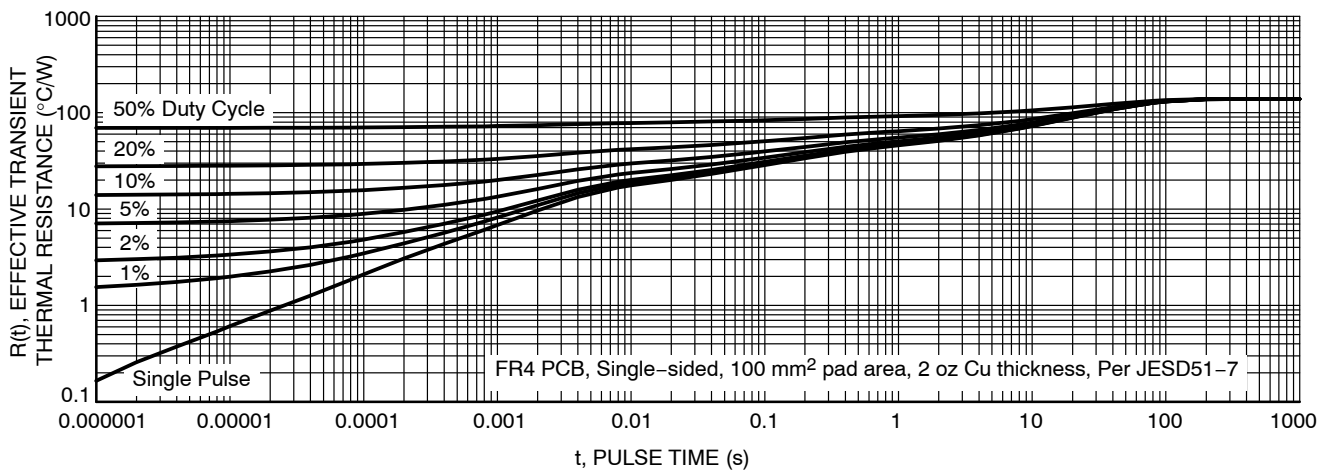


Figure 10. Transient Thermal Impedance from Junction-to-Ambient as a Function of Pulse Duration

BCP53M

ORDERING INFORMATION

Device	Marking	Package	Shipping [†]
BCP53MTWG	3M	WDFNW3 (Pb-Free)	3000 / Tape & Reel
BCP5310MTWG	3N		
BCP5316MTWG	3P		
NSVBCP53MTWG*	3M		
NSVBCP5310MTWG*	3N		
NSVBCP5316MTWG*	3P		

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

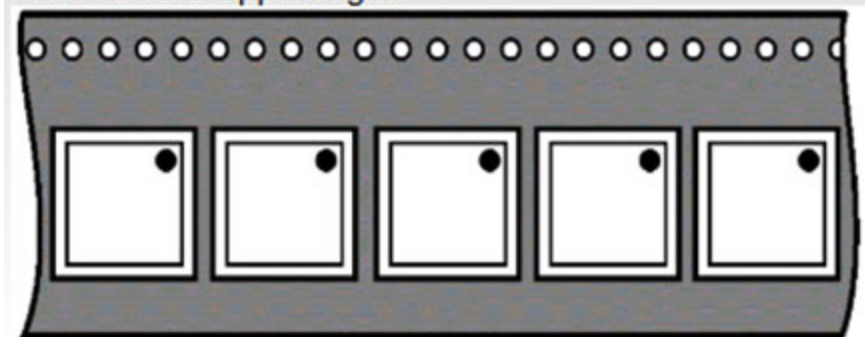
*NSV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q100 Qualified and PPAP Capable.

Pin 1 Orientation in Tape and Reel

Direction of Feed

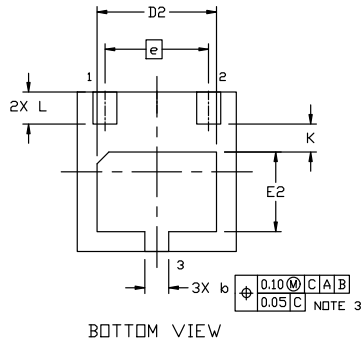
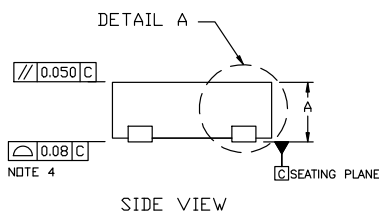
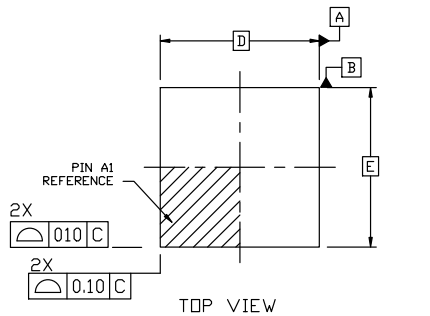
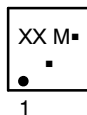


Pin 1 Toward Upper Right




WDFNW3 2x2, 1.3P
CASE 515AA
ISSUE A

DATE 26 JUL 2022


GENERIC
MARKING DIAGRAM*


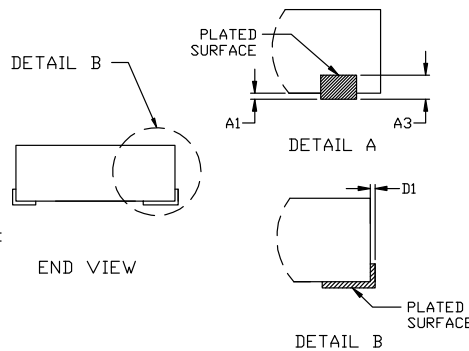
XX = Specific Device Code
M = Date Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

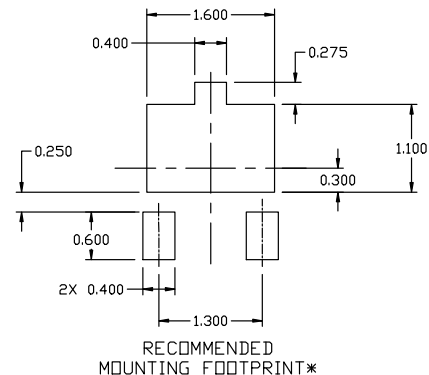
*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
2. CONTROLLING DIMENSION: MILLIMETERS
3. DIMENSION b APPLIES TO PLATED TERMINAL AND IS MEASURED BETWEEN 0.15 AND 0.30MM FROM THE TERMINAL TIP.
4. COPLANARITY APPLIES TO THE EXPOSED PADS AS WELL AS THE TERMINALS.



DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.70	0.75	0.80
A1	0.00	---	0.05
A3	0.20 REF		
b	0.25	0.30	0.35
D	2.00 BSC		
D1	0.00	---	0.04
D2	1.40	1.50	1.60
E	2.00 BSC		
E2	0.90	1.00	1.10
e	1.30 BSC		
K	0.35 REF		
L	0.35	0.40	0.45



* For additional information on our Pb-Free strategy and soldering details, please download the DN Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

DOCUMENT NUMBER: 98AON33309H

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DESCRIPTION: WDFNW3 2x2, 1.3P

PAGE 1 OF 1

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